

- Provides all computer vision functions
- Used for: video reading, image processing, drawing, display

Line 2: import math

- **math** = Python's built-in math library
- We need it for: π (pi) in circularity formula
- Formula: $4 \times \pi \times \text{area} / \text{perimeter}^2$

What These Libraries Provide

Library	Functions We Use	Purpose
cv2	VideoCapture, cvtColor, GaussianBlur, Canny, findContours, circle, putText, imshow	All video and image operations
math	math.pi	Mathematical constant $\pi \approx 3.14159$

3 Section: Video Loading

```
cap = cv2.VideoCapture("video.mp4") if not cap.isOpened():
print("Error: Video could not be opened!") exit() else: print("Video
loaded successfully!")
```

Line 3: Load Video

```
cap = cv2.VideoCapture("video.mp4")
```

What it does: